

REX LU · TAIPEI

I design the architecture, and run an AI agent pipeline to ship it.

16 years in systems development and architecture, specializing in payment security, high-concurrency platforms and multi-tenant systems. I now work as an agentic engineer: I own the architecture, task breakdown and quality gates, while parallel AI agents handle implementation — one person delivering what used to take a team.

```
~/projects · zsh

$ git log --author=rex --since=2025-10
--no-merges | wc -l
3,300+
$ claude --worktrees --parallel
✓ 44 worktrees · cross-model review on
$ report --month=2026-08
points 743  team share 60%  2.9× the #2
$ ls ~/Applications/home-cooked
MyGhost  MyXnip  TypeTop  AirGrip  +8
$ _
```

Systems Architect · Agentic Engineer · Toolsmith

ABOUT

Understand the business logic first, then make the system solid

I've been building websites since 2010. Early roles at Yahoo! Super Mall, Wretch and PChome mobile payments (lead developer on the first version of Pi Wallet) gave me a foundation in e-commerce and payments. I then led overseas gaming platform teams in the Philippines and Taiwan — up to 21 engineers at once — running online platforms with weekly revenue above NT$100 million.

What I care about most is that the money is never wrong. Cross-system payment confirmation, timeouts and duplicate credits, reconciliation and recovery — I design all of these in from day one; handling the failure cases is often more than three times harder than the happy path. My recent work on two-phase transfer safety, withdrawal dispatch guards and load-test tuning all comes from this principle.

Since 2025 I've switched entirely to agentic engineering: Claude Code, Codex and Gemini form a reviewable, re-runnable development pipeline, while I focus on architecture, task breakdown, review and release gating. A small team can deliver what a large one does.

2.9×	62.8→0.03%	90s→2s	-96%	75 in 2 days	7 days
One month's output vs. the team's #2	Failure rate in a production-scale load test	Query on a 13.47M-row table	SQL statements for a 30-platform report	Third-party platform configs auto-extracted	AI product from kickoff to public beta

AGENTIC ENGINEERING

Not just using AI — running agents as a production line

In 2026 the industry calls this way of working agentic engineering: humans own architecture and quality, while AI agents write, test and ship. This is the pipeline I actually run every day.

01

Architect & break down

Pin down business rules and failure cases first, split requirements into independently verifiable units, and write the acceptance criteria.

`Owned by me`

02

Parallel implementation

One git worktree per ticket, so many agents develop at once without stepping on each other — up to 44 parallel workspaces on a single project.

`Run by AI agents`

03

Cross-review & diff testing

Claude writes, Codex reviews. Rewrites treat the legacy system's real output as ground truth, verified column by column with a custom black-box diff tool.

`Cross-model + tooling`

04

Gate & release

Human review, tests and canary flags before anything reaches production; the process is captured as re-runnable skills and docs.

`Owned by me`

Proof: one month of a shrinking team — in August 2026, three teammates left one after another and the team was down to 4 active developers.

- Delivered 60% of the team's story points and 56% of merged MRs single-handedly, across five tracks: backend, admin, frontend, store and cloud
- Integrated 6 game providers in one week, each through a three-stage rollout with live testing
- Fixed 12 production ops errors in the week before launch and ran the three-environment release solo
- Wrote 31 assessments, specs and handover documents the same month — about 220,000 Chinese characters

RECENT WORK · 2025–2026

Representative projects from the past year

Client projects are described by product type for confidentiality.

FULL-TIME

Game API aggregation platform: risk control & new-market wallet

2020.04 – present

Lead developer for risk control and the merchant back office. Built an SMS-invite marketing system from scratch (Cloudflare Workers + Analytics Engine), a time-zone architecture spanning 8 time zones × 4 systems, and the internationalization of a trilingual South Asian wallet.

Large-table query 90s → 2–3s · new system live in one month

PHP · ThinkPHP · Cloudflare Workers · Redis · Nuxt

CONTRACT

White-label entertainment platform: wallet, payments & cloud

2026.07 – present

Two-phase transfers with timeout reconciliation and withdrawal dispatch guards; WAF allowlists pushed directly by the app and written back to Terraform; a production-scale load test surfaced three root causes. 788 commits in three months — 29% of the team's total.

Failure rate 62.8% → 0.03% · p95 60s → 10.9s

Go · AWS EKS · WAF · Terraform · Helm · Vue 3

Multi-tenant platform: affiliate commission back-office API rewrite

2025.10 – 2026.09

Rewrote legacy PHP into a three-layer Lumen 10 architecture, writing 73 of the 86 routes; report aggregation across 70+ schemas; config-driven support for 75 platforms at once; a declarative diff-sync engine that passed client acceptance.

SQL statements -96% · 41 of 42 platforms match old vs. new column by column

PHP 8.3 · Lumen · PHPUnit · MySQL · Redis

Contractor payment slips & general ledger system

2026.03 – present

Moved contractor payment slips online for three companies in a group: a tax-rule engine (including the 183-day residency rule), four-level approvals, 20,000-row batch uploads and bilingual PDFs. The follow-on accounting system has finished requirements interviews and a four-phase plan.

3 APIs + 3 frontends · delivered and still growing

PHP 8 · Next.js 16 · TypeScript · MariaDB

Full-stack platform for an astrology brand

2026.04 – present

Content site, membership and interactive divination tools, built solo. AI drafts, experts review before publishing, and the generation worker is sandboxed. After an SEO/AEO overhaul the core keyword ranks #1 on Bing and Google impressions rose 58%.

121 APIs · zero-downtime domain migration

Next.js · PHP · Three.js · Gemini ⇄ Claude · LINE Login

OWN PRODUCTS

NianNian: AI character companion platform

2026.09

Layered memory, proactive messages driven by real presence heartbeats, a 6-stage relationship arc and multi-character group chat. Conversations stream through Gemini while other generation runs on a local Claude to keep costs down.

7 days from kickoff to public beta

Node.js · SQLite · SSE · Gemini · Claude CLI

Rankly: industry website rankings & AI visibility

2026.08 – present

Daily health checks on clients' and competitors' sites, plus real questions put to ChatGPT, Claude, Gemini and Perplexity to rank who AI actually sees; includes a five-point GEO diagnosis and daily content trends.

857 AI answers in 45 days · found the bottleneck is coverage, not rank

Node.js · Multi-LLM · headless Claude · GEO

TOOLSMITH · HOME-COOKED SOFTWARE

I build my own tools

The ideal team in *The Mythical Man-Month* includes a "toolsmith" who builds the tools everyone else relies on. AI agents make that cheap: rather than living with the limits of commercial software, build one that fits your own workflow. In 2026 I built 12 native macOS apps, and I use every one of them daily. Public ones link to their GitHub downloads.

Customized Ghostty fork

MyGhost

A terminal built for long-running AI agents: sidebar tabs, tmux so sessions survive crashes, auto-reconnecting remote SSH tabs, pasted images auto-uploaded to remote agents, a Claude/Codex quota dashboard, plus a mobile-friendly web version.

Download latest on GitHub →

Own build, Xnip-style UI

MyXnip

The old screenshot API could freeze the whole machine, so I redesigned it on ScreenCaptureKit as capture first, display later. Scrolling captures, annotation, OCR.

Rewrite of an existing app

TypeTop

Hold right ⌘ and talk: Whisper transcribes, an LLM cleans it up and types it into any text field. Supports on-device Apple Intelligence and automatic failover across 7 LLM providers.

Download latest on GitHub →

Own build, modeled on Going IME

SuperNatural IME

A custom Chinese input engine on InputMethodKit: DP word-lattice composition, selection learning with decay, abbreviation shortcuts.

Original

AirGrip

Control your Mac with hand gestures through the camera, using Vision hand-pose detection — all on-device.

Customized DuoBar fork

MyDuoBar

Menu-bar status hub: Wi-Fi, VPN, per-earbud AirPods battery and the native charge limit, under Swift 6 strict concurrency.

Download latest on GitHub →

Own build, Downie-style UI

My Downie

Video downloader that covers platforms the original can't; one-click launch from a browser extension, and the app updates itself.

Own build, Niya-style UI

My Niya

A Dynamic Island-style dashboard at the top of the screen: system monitor, to-dos, network speed, clipboard and AirDrop.

Own build, inspired by Hidden Bar

MyHiddenBar

Worked out the new menu-bar layout rules in macOS 27 and rewrote the hiding logic — done in a day.

Own build, Obsidian-style UI

MyObsidian

Markdown knowledge base: backlinks, mermaid, cross-platform builds and e2e tests.

Download latest on GitHub →

Own build, Slidepad-style UI

MySlidePad

A browser panel that slides out when you touch the screen edge, with scripts for organizing AI chats.

Download latest on GitHub →

Original

MyPlayer

A lightweight player with keyboard control of speed and looping.

SKILLS

Skills

AI & Agentic

Claude Code / Codex / Gemini

Parallel multi-agent dev on worktrees

Cross-model review · reusable skills

LLM product integration & cost routing

GEO / AEO

Backend & Data

PHP (Laravel · Lumen · ThinkPHP)

Go (Gin · GORM)

Node.js

MySQL / MariaDB tuning · Redis

Queues · scheduling · idempotency

Frontend & Client

Next.js · Nuxt · Vue 3

TypeScript · Tailwind

Swift · SwiftUI · AppKit

PWA · Three.js

i18n

Cloud & Architecture

AWS EKS · WAF · CloudFront · RDS

Terraform · Helm · GitLab CI

Cloudflare Workers · Tunnel

High concurrency · load testing · canary

Payment security & reconciliation

EXPERIENCE

Experience

2020.04 – present

Overseas game API aggregation platform

Engineering Team Leader • Taipei

Extended Laravel and ThinkPHP systems and led R&D in building the collection-merchant architecture and operations back office; integrated third-party payments including Alipay, bank cards, mobile top-up and USDT; led cross-border communication (Philippines, Dubai) and remote collaboration. Recently responsible for risk control, the merchant back office, marketing systems and new-market wallets.

2025 – present

Contract development

Systems architecture • Full-stack • Agentic engineering

Enterprise platform rewrites and embedded development (affiliate commission back office, wallet payments and cloud), plus from-scratch delivery for SMBs and brands (contractor payment & accounting system, astrology brand platform).

2019.05 – 2020.01

Taiwan gaming fintech company

Engineering Team Leader • Taipei

Built third-party game platform integrations in Yii and Go, owning game logic and financial reporting; developed a real-time routing and hedging system for large orders in Go.

2017.07 – 2019.01

Desheng Group (Philippines)

Deputy Director of Engineering • Taipei

Sent from the Manila headquarters to build a team in Taiwan, leading up to 21 people. Built a user analytics center, game center, top-up, risk control and revenue-sharing system on Laravel; led the team at G2E Asia 2018 in Macau.

2016.04 – 2017.06

GamePark Group (Philippines)

Team Leader • Manila

Ran online platforms averaging over NT$100 million in weekly revenue: built one new platform and operated four; shared multi-platform core, NoSQL load testing and mobile game APIs.

2014.11 – 2016.04

PChome Online

Team Leader • Taipei

Lead developer on the first version of the Pi mobile wallet: multi-host architecture, a database caching package, bank payments, risk control and accounting; introduced testing and Scrum.

2010 – 2014

Yahoo! • Tripmoment • Interfly

Web Development Engineer • Taipei

Purchase data analysis and ad partnerships for Yahoo! Super Mall; refactored Wretch (test coverage 70%→80%); app APIs for a travel startup; a government website that won Taiwan's Golden Finger Award.

CONTACT

Need a system made solid, or want AI in your workflow?

Full-time roles, consulting and contract work are all welcome — just email me. Online: resume-rex.master0987.com/en

lance70176@gmail.com

GitHub